Speech-to-Text (STT) Module Documentation

Objective

Convert extracted audio files into text transcripts that can be used as input for the Retrieval-Augmented Generation (RAG) pipeline.

Implementation

OpenAI Whisper was used to convert audio into text. Initially, the large-v2 model was tested because it provides the highest transcription accuracy. However, it was extremely slow on a CPU-only system. Therefore, the base model was selected because it offers a good balance between transcription quality and execution speed while producing reliable results.

Optimization

To speed up development and testing, a 10-second sample audio file was created using FFmpeg. Processing the sample audio allowed the Speech-to-Text pipeline to be tested quickly without waiting for the complete audio file to finish processing.

JSON Output Processing

Whisper returns the transcription output as a Python dictionary containing the transcript, language information, timestamps, and speech segments. The output is saved into a JSON file using Python's json module. The JSON file is then filtered so that only the required information is retained for the next stage of the RAG pipeline.

Language Translation

The source videos are primarily in Hindi. Whisper is configured using the translate task, allowing Hindi speech to be converted directly into English text. This English transcript is used for text chunking, embedding generation, vector database storage, and retrieval.

Workflow

Video → FFmpeg → MP3 → Whisper (Base Model) → English Transcript → JSON Output → Filtered JSON → Chunking → Embeddings → Vector Database → RAG

Conclusion

The Speech-to-Text module is an important component of the RAG system. FFmpeg extracts audio from videos, while OpenAI Whisper converts Hindi speech into English text. The processed JSON output provides clean textual data for the downstream RAG pipeline.